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1. Angular is a Type-Script based open source front-end application framework developed by google.

It is used to build dynamic, single page web application by extending HTML Syntax and manage web components.

Primary Purpose:

→ Simply web development & testing web applications.

→ allow Component based development, making it scalable and maintainable.

2. Main features of Angular.

→ Component Based Architecture:

Application are built using Component.

→ Two way Data Binding: Automatically synchronize data between the model and view.

→ Dependency Injection: Manages service instance efficiently.

→ TypeScript Support: Static typing and modern JavaScript features.

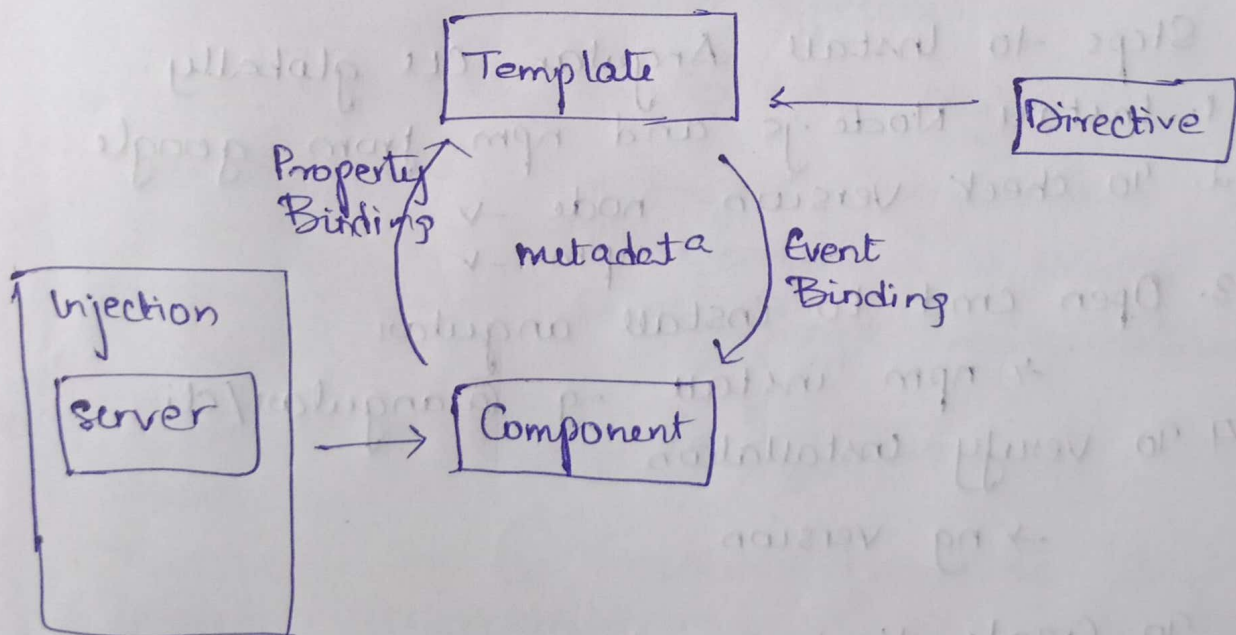
→ CLI Tool: Angular CLI for rapid development and configuration.

→ Cross-Platform: Support mobile, desktop and web applications.

3. Difference of Angular 2 way Binding and React 1 way data flow.

- Angular 2 way Binding - data flows in both direction from Component to view and from view to Component
- when the input is updated the model Component is updated automatically, when model changes the view updates too.
- React 1 way Data flow
- Data flow in 1 direction - from parent to Child Component and from Component state to view.
- we use onchange handler to manually update the state.

4. Basic Architecture of Angular Application with a diagram.



Component Building block of Angular application
Template user interface of view of user
Directive instruction of Dom
Service are used when data & logic need
to used in other Components
Injection helps in Code reusability & testing

5. Ngmodule is a decorator represented as @NgModule that defines a compilation context for Component.

It helps group Component, directives and Services.

It declares what Component belong to
Import necessary Angular modules

Define root Component via bootstrap
Provide Service using providers array

6. Angular CLI is a tool that create an angular application

Serve the application in development mode.

Generate Component, Service, modules.

Build and deploy Angular Apps.

It automates Configuration and Saves development time.

7. Steps to install Angular CLI globally.

1. Install Node.js and npm from google

2. To check version node -v
npm -v

3. Open cmd to install angular

→ npm install -g @angular/cli

4. To verify installation

→ ng version

8. To Create first angular application using Angular cli

1. Install the angular cli

npm install -g @angular/cli

2. To create project

ng new projectname

3. Now to change the path into the project folder

cd projectname

4. To run/start it
ng serve

9. Default directories that are created while creating a new project

Projectname

↳ Src

↳ app/

↳ Component/modules

← Main Application folder

Development/production

index.html

← To HTML file

main.ts

← To Typescript

Server.ts

Style.css

← To add styles

angular.json

→ Project Configuration

package.json

→ Dependencies & Scripts

tsconfig.json

→ TypeScript Configuration

10. TypeScript is a Superset of JavaScript with static typing.

- TypeScript reduces runtime error as it is compile time error finder

- AutoCompletion & Tooling Support
- Class based oops features

- Decorators for metadata (@Component, @NgModule)

- Easier large-scale application & Data type finder

11. TypeScript enhance the development process in Angular by :

Early error detection : Type checking at Compile time

Improved IDE Support : Code Completion hints , Documentation

Async Programming Support : Built in Support for promises and async/await.
Safe Variable and method renaming.